UK Housing Affordability Intelligence API (1.0.0)

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URL: https://github.com/AlexJGeorge1/COMP3011-housing-api | License: MIT

Combines HM Land Registry Price Paid Data, ONS Annual Survey of Hours and Earnings, and ONS Private Rental Market Summary to provide affordability insights across UK regions.

Health

Service health check

Returns a simple status message confirming the API is running.

Responses

> **200** Successful Response

GET /health

Response samples

200

Content type
application/json

Copy

```
null
```

Authentication

Login For Access Token

OAuth2 compatible token login, required for Swagger UI authentication. Uses demo credentials from config since this is a public dataset API with writing restricted to an admin.

REQUEST BODY SCHEMA: application/x-www-form-urlencoded
required

`grant_type` >	Grant Type (string) or Grant Type (null) (Grant Type)
`username` required	string (Username)
`password` required	string (Password)
`scope`	string (Scope) Default: `" "`
`client_id` >	Client Id (string) or Client Id (null) (Client Id)
`client_secret` >	Client Secret (string) or Client Secret (null) (Client Secret)

Responses

> **200** Successful Response

> **422** Validation Error

POST `/token`

Response samples

200 422

Content type
application/json

Copy

```
null
```

Listings

List all property listings

QUERY PARAMETERS

`region` >	Region (string) or Region (null) (Region) Filter by region name
`limit`	integer (Limit) <= 500 Default: `100` Maximum number of results to return
`offset`	integer (Offset) Default: `0` Number of results to skip (for pagination)

Responses

> **200** Successful Response

> **422** Validation Error

GET /listings

Response samples

200 422

Content type

application/json

Copy Expand all Collapse all


```
[
 - {
        "address": "string",
        "region": "string",
        "price": 0,
        "bedrooms": 0,
        "property_type": "string",
        "transaction_date": "2019-08-24",
        "id": "string",
        "created_at": "2019-08-24T14:15:22Z"
    }
]
```


Create a new listing

AUTHORIZATIONS: > *OAuth2PasswordBearer*

REQUEST BODY SCHEMA: application/json
required

address required	string (Address) Full property address
region required	string (Region) ONS region name
price required	integer (Price) Sale price in GBP
bedrooms >	Bedrooms (integer) or Bedrooms (null) (Bedrooms) Number of bedrooms
property_type >	Property Type (string) or Property Type (null) (Property Type) e.g. detached, semi, terraced, flat
transaction_date required	string <date> (Transaction Date) Date the transaction completed

Responses

> **201** Successful Response

> **422** Validation Error

POST /listings

Request samples

Payload

Content type
application/json

Copy

```
{
    "address": "string",
    "region": "string",
    "price": 0,
    "bedrooms": 0,
    "property_type": "string",
    "transaction_date": "2019-08-24"
}
```

Response samples

201 **422**

Content type
application/json

Copy

```
{
    "address": "string",
    "region": "string",
    "price": 0,
    "bedrooms": 0,
    "property_type": "string",
    "transaction_date": "2019-08-24",
    "id": "string",
    "created_at": "2019-08-24T14:15:22Z"
}
```

Get a single listing by ID

Returns one property listing

PATH PARAMETERS

`listing_id` required	string (Listing Id)

Responses

> **200** Successful Response

> **422** Validation Error

GET `/listings/{listing_id}`

Response samples

200 | 422

Content type

application/json

Copy

```
{
    "address": "string",
    "region": "string",
    "price": 0,
    "bedrooms": 0,
    "property_type": "string",
    "transaction_date": "2019-08-24",
    "id": "string",
    "created_at": "2019-08-24T14:15:22Z"
}
```

Update a listing

AUTHORIZATIONS: > *OAuth2PasswordBearer*

PATH PARAMETERS

listing_id required	string (Listing Id)

REQUEST BODY SCHEMA: application/json
required

address >	Address (string) or Address (null) (Address)
region >	Region (string) or Region (null) (Region)
price >	Price (integer) or Price (null) (Price)
bedrooms >	Bedrooms (integer) or Bedrooms (null) (Bedrooms)
property_type >	Property Type (string) or Property Type (null) (Property Type)
transaction_date >	Transaction Date (string) or Transaction Date (null) (Transaction Date)

Responses

> **200** Successful Response

> **422** Validation Error

PUT /listings/{listing_id}

Request samples

Payload

Content type
application/json

Copy

```
{
    "address": "string",
    "region": "string",
    "price": 0,
    "bedrooms": 0,
    "property_type": "string",
    "transaction_date": "2019-08-24"
}
```

Response samples

200 422

Content type

application/json

Copy


```
{
    "address": "string",
    "region": "string",
    "price": 0,
    "bedrooms": 0,
    "property_type": "string",
    "transaction_date": "2019-08-24",
    "id": "string",
    "created_at": "2019-08-24T14:15:22Z"
}
```


Delete a listing

AUTHORIZATIONS: > *OAuth2PasswordBearer*

PATH PARAMETERS

`listing_id` required	string (Listing Id)

Responses

— **204** Successful Response

> **422** Validation Error

DELETE `/listings/{listing_id}`

Response samples

422

Content type

application/json

Copy Expand all Collapse all


```
{
  - "detail": [
      + { … }
    ]
}
```


Regions

List all regions

Responses

> **200** Successful Response

GET /regions

Response samples

200

Content type

application/json

Copy Expand all Collapse all

```
[
  - {
        "name": "string",
        "ons_code": "string",
        "median_salary": 0,
        "median_rent": 0,
        "year": 0,
        "id": "string"
    }
]
```


Create a new region

AUTHORIZATIONS: > *OAuth2PasswordBearer*

REQUEST BODY SCHEMA: application/json
required

`name` required	string (Name) Region name
`ons_code` required	string (Ons Code) ONS area code
`median_salary` >	Median Salary (number) or Median Salary (null) (Median Salary) Median salary in GBP/year from ONS ASHE
`median_rent` >	Median Rent (number) or Median Rent (null) (Median Rent) Median rent in GBP/month from ONS PRS
`year` required	integer (Year) Data year

Responses

> **201** Successful Response

> **422** Validation Error

POST /regions

Request samples

Payload

Content type

application/json

Copy

```
{
    "name": "string",
    "ons_code": "string",
    "median_salary": 0,
    "median_rent": 0,
    "year": 0
}
```

Response samples

201 **422**

Content type

application/json

Copy

```
{
    "name": "string",
    "ons_code": "string",
    "median_salary": 0,
    "median_rent": 0,
    "year": 0,
    "id": "string"
}
```

Get a single region by ID

PATH PARAMETERS

`region_id` required	string (Region Id)

Responses

> **200** Successful Response

> **422** Validation Error

GET /regions/{region_id}

Response samples

200 422

Content type
application/json

Copy

```
{
    "name": "string",
    "ons_code": "string",
    "median_salary": 0,
    "median_rent": 0,
    "year": 0,
    "id": "string"
}
```

Update a region

AUTHORIZATIONS: > *OAuth2PasswordBearer*

PATH PARAMETERS

`region_id` required	string (Region Id)

REQUEST BODY SCHEMA: application/json
required

`name` >	Name (string) or Name (null) (Name)
`ons_code` >	Ons Code (string) or Ons Code (null) (Ons Code)
`median_salary` >	Median Salary (number) or Median Salary (null) (Median Salary)
`median_rent` >	Median Rent (number) or Median Rent (null) (Median Rent)
`year` >	Year (integer) or Year (null) (Year)

Responses

> **200** Successful Response

> **422** Validation Error

PUT /regions/{region_id}

Request samples

Payload

Content type

application/json

Copy

```
{
    "name": "string",
    "ons_code": "string",
    "median_salary": 0,
    "median_rent": 0,
    "year": 0
}
```

Response samples

200 **422**

Content type

application/json

Copy

```
{
    "name": "string",
    "ons_code": "string",
    "median_salary": 0,
    "median_rent": 0,
    "year": 0,
    "id": "string"
}
```

Delete a region

AUTHORIZATIONS: > *OAuth2PasswordBearer*

PATH PARAMETERS

`region_id` required	string (Region Id)

Responses

— **204** Successful Response

> **422** Validation Error

DELETE /regions/{region_id}

Response samples

422

Content type
application/json

Copy Expand all Collapse all

```
{
  - "detail": [
      + { … }
    ]
}
```

Analytics

Housing affordability metrics for a region

PATH PARAMETERS

region_name required	string (Region Name)

Responses

> **200** Successful Response

> **422** Validation Error

GET /affordability/{region_name}

Response samples

200 422

Content type

application/json

Copy

```
{
    "region": "string",
    "ons_code": "string",
    "year": 0,
    "median_salary": 0,
    "median_rent_monthly": 0,
    "median_house_price": 0,
    "price_to_income_ratio": 0,
    "rent_to_income_pct": 0,
    "affordability_band": "string"
}
```

Year-on-year house price trends for a region

PATH PARAMETERS

region_name required	string (Region Name)

Responses

> **200** Successful Response

> **422** Validation Error

GET /trends/{region_name}

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all


```
{
    "region": "string",
    "data_from": 0,
    "data_to": 0,
  - "yearly_data": [
      + { … }
    ],
    "total_change_pct": 0,
    "cagr_pct": 0
}
```


Rent-vs-buy comparison for a region

Compares estimated monthly mortgage cost against median rent, and estimates how many months a median-salary earner would need to save a 10% deposit.

Mortgage assumptions: 10% deposit, 4.5% p.a. interest (approx. Bank of England base rate + typical lender margin), 25-year repayment term.

PATH PARAMETERS

region_name required	string (Region Name)

Responses

> **200** Successful Response

> **422** Validation Error

GET `/rent-to-buy/{region_name}`

Response samples

200 | 422

Content type

application/json

Copy

```
{
    "region": "string",
    "median_house_price": 0,
    "monthly_mortgage_estimate": 0,
    "monthly_rent": 0,
    "monthly_saving_vs_renting": 0,
    "deposit_10pct": 0,
    "months_to_save_deposit": 0,
    "years_to_save_deposit": 0,
    "verdict": "string"
}
```

Intelligence

Semantic search over property listings

QUERY PARAMETERS

q required	string (Q) >= 3 characters Natural language search query, e.g. 'affordable flat in London'

`limit`	integer (Limit) `<= 50` Default: `10` Maximum number of results
`region` >	Region (string) or Region (null) (Region) Optional region filter

Responses

> **200** Successful Response

> **422** Validation Error

GET /search

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all

```
{
    "query": "string",
  - "results": [
      + { … }
    ],
    "count": 0
}
```

AI-generated housing market intelligence for a region

PATH PARAMETERS

`region_name` required	string (Region Name)

Responses

> **200** Successful Response

> **422** Validation Error

GET /insights/{region_name}

Response samples

200 422

Content type

application/json

Copy Expand all Collapse all


```
{
    "region": "string",
    "insight": "string",
    "data_used": { },
    "powered_by": "string"
}
```